

Argmax Decoding Causes Liquidity Failure in a Market of Language-Model Traders

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September 2026

Abstract

We populate a continuous double auction with traders whose decisions are made by a typed language model that returns a probability distribution over actions, and vary only how that distribution is consumed. Taking the model’s modal action—a common deployment choice—makes the market stop trading after shocks: following a positive shock, the market clears in 65.6% of periods against 91.3% after a negative shock, a gap of 25.8 percentage points (95% CI [17.3, 34.2], $t = 6.37$, $n = 20$, preregistered). Sampling from the same distribution reduces the gap to 3.1 points. Two algorithmic baselines that are symmetric by construction show no gap.

The mechanism is a single one. Every trader conditions on the same public signal, so a deterministic decoder makes them act identically, and a market of unanimous buyers has no counterparty. Measured at the shock itself, 90.0% of traders take the same side after a positive shock against 59.4% after a negative one, and 100% of non-trading periods under modal decoding had one side of the order book entirely empty, against 3.4% for the zero-intelligence baseline.

We test that mechanism with a preregistered prediction it could have failed: if unanimity is the cause, the effect must weaken sharply as the market grows. It does, from 37.0 points at four traders to 4.2 at thirty-two (-0.112 per doubling, seed-paired $t = -7.78$). A companion prediction—that sampling would show no such decay—is *unresolved*: the seed-paired estimate is $t = -2.00$, two-sided $p = 0.060$, which neither falsifies it nor, absent a preregistered equivalence margin, confirms it.

Separately, we find that ordinary domain wording induces a 25-point asymmetry in the model’s stated conviction between buying and selling at equal mispricing, which vanishes under mirror-image phrasing. A preregistered test found this does *not* move pricing error, and we report that null.

Every confirmatory hypothesis test was preregistered on seeds disjoint from those that generated it; the wording study and the order-book diagnostics are exploratory and labelled as such. Of seven registered predictions, three were confirmed (D1, D2, E1), three failed (C1, C2, and the original clause that modal decoding would be immune) and one is unresolved (E2); all are reported. A separate post-hoc mechanism account is withdrawn and identified as such.

1 Introduction

A language model asked to choose among actions returns, in principle, a distribution. A common deployment choice is to discard it and take the mode. For a single decision that selects the most probable action, which is the right choice under 0–1 loss against the returned distribution. This

paper asks what happens when many agents driven by the same model make related decisions at the same time.

The answer, in a market, is that the market stops. Modal decoding makes agents conditioning on shared information act identically. In a setting that requires counterparties, unanimity is not consensus; it is the absence of trade. We measure this in a continuous double auction, isolate the mechanism, and confirm it with a prediction that could have failed and did not.

Our contributions:

1. **A liquidity failure caused by decoding, not by reasoning.** The model’s modal action is *correct* on both sides at every mispricing in our static probe grid, and at the shock itself in the market it takes the correct side in both directions. The failure is in the aggregation of decisions that are individually defensible.
2. **A mechanism measured in the market.** We instrument both sides of the order book and show that under modal decoding, non-trading periods lack a counterparty essentially always, while for symmetric baselines they do not. Agreement at the shock is itself directional—90.0% after a positive shock against 59.4% after a negative one—so unanimity accounts for both the magnitude of the halt and which shock it strikes. An earlier draft posited a second component; Section 6 explains why that was an artefact of reading a static probe as a market statistic, and withdraws it.
3. **A falsifiable size-scaling test.** The mechanism implies the effect must decay with market size at a rate set by the agreement rate. We registered this before running it; it held.
4. **A prompt-sensitivity result, and a null.** Domain-idiomatic wording induces a large conviction asymmetry invisible to anyone reading only the modal action. A preregistered test found it does not move pricing error.

We emphasise the failures, and state the registration record exactly. **Three preregistered predictions failed:** C1, that the wording asymmetry would move pricing error (a clean null, Section 5); the original clause that modal decoding would be immune to the liquidity effect; and C2, that sampling would show the *larger* wording contrast, on which the estimate came back the wrong sign ($+0.161$, $p = 0.80$). **One is unresolved:** E2 (Section 7). **Three were confirmed:** D1, D2 and E1.

Separately, and not part of the registration record, we withdraw a *post-hoc* two-component account of the mechanism that an earlier draft proposed (Section 6). The preregistration for that run deliberately instrumented rather than predicted, so this was never a registered hypothesis and we do not count it as one. The result that survived did so against a test designed to kill it.

2 Related work

Market design and baselines. The experimental market tradition begins with Smith [1962]. Our lower baseline is the zero-intelligence trader of Gode and Sunder [1993], whose result—that budget-constrained random traders achieve high allocative efficiency—makes a random arm a genuine floor rather than a strawman, and motivates the value clamp that prevents any arm in our design from knowingly trading at a loss.

LLM agents in simulated markets. Lopez-Lira [2025] builds a simulated stock market with a persistent order book in which heterogeneous LLM agents submit structured decisions, and reports that agents adhere consistently to assigned strategies while the market reproduces price discovery, bubbles and underreaction. That work also notes that prompts can generate correlated behaviours affecting market stability, which is the phenomenon we isolate and measure here.

Correlation as systemic risk. Closest to our result is Ross et al. [2026], who argue that shared training and architecture make more capable models behave more similarly, creating a non-diversifiable risk floor, and who test this with LLM traders of varying capability in an agent-based market. They find correlated behaviour that *increases* with capability, and that it becomes a liability when agents share a misinformation environment.

Our contribution is complementary and, we think, orthogonal in a useful way. They vary the model and attribute correlation to properties of the model—its training, its architecture, its capability. We hold the model *fixed* and vary only how its output distribution is consumed. The correlation we measure is therefore not solely a property of the model but also of a deployment choice, and that component a practitioner controls directly: our argmax and sampling arms issue identical requests and read identical responses, differing only in a single line of client code. Where Ross et al. [2026] identify a risk floor set by model similarity, we show a large component of realised correlation sitting above that floor, substantially reduced here without changing models—though sampling still leaves a 3.1-point gap, so the decoder does not account for all of it.

Herding and cascades. The classical account of correlated action is the informational cascade of Bikhchandani et al. [1992], in which an agent rationally ignores private information after observing enough predecessors. Our mechanism is deliberately simpler and should not be confused with it. Agents in our market do not observe one another’s decisions and draw no inference from them; they condition on a common public signal and a deterministic policy maps that signal to the same action. There is no learning, no sequence, and no inference—only a degenerate policy. The resulting unanimity looks like herding and has the same market consequence, but it arises without any of the cascade machinery, which is why it can be switched off at the decoder.

Decoding and diversity. That greedy decoding collapses output diversity is long established in text generation: Holtzman et al. [2020] show that maximisation-based decoding yields repetitive, degenerate text and introduce nucleus sampling as a remedy. The quality–diversity tension is well understood *within* a single generation. We report a consequence one level up: when many agents share a decoder, the diversity that sampling preserves is not a stylistic property of one output but the raw material a market needs to clear. Sampling is individually worse and collectively necessary.

Confidence and calibration. Kadavath et al. [2022] study whether language models can predict the correctness of their own answers, and subsequent work has documented systematic overconfidence in verbalised confidence. Our Experiment 5 touches this literature from an unusual angle: the model we use returns a probability distribution natively rather than a verbalised estimate, and we find that ordinary domain wording shifts that distribution by 25 points without changing the selected option. Any calibration measurement taken over such a model inherits whatever asymmetry the question’s phrasing introduced.

What we could not find. We did not find prior work isolating the decode rule as a cause of market-level liquidity failure, nor a preregistered size-scaling test of a herding mechanism in an LLM-populated market. We would welcome pointers; this survey was conducted by the authors and is not exhaustive.

3 The instrument

3.1 Market

A continuous double auction in one abstract good. Prices are integer ticks; matching is price-then-time priority; an incoming order executes at the resting order’s price. A trader’s resting orders are withdrawn at the top of its turn, so each trader has at most one live order. Submitting on one side cancels the trader’s resting orders on the other, preventing wash trades without leaving a crossed book.

Accounts are conserved by construction. Each trade moves cash from buyer to seller and inventory from seller to buyer, one for one. Short sales and borrowing are disallowed, and every order is pre-checked against uncommitted cash or uncommitted inventory. A property test submits 20,000 random orders and asserts after every one that total cash and total units are exactly their endowed values, that no account is negative, that no trader has committed more than it holds, and that the book is never crossed.

Order size is fixed at one unit, so aggressiveness is the only lever an arm has; position sizing is a second channel and is deliberately disabled.

3.2 Fundamental and information

The fundamental F_t is piecewise constant. For the size and liquidity experiments we use a *matched-jump* process: jumps are evenly spaced, of identical magnitude, strictly alternating in direction, and even in number, so that every up window is paired with a down window of the same size between the same two price levels. This makes the up-versus-down comparison within-run and matched, which matters: under a random-jump process the per-run noise on this statistic has standard deviation 0.74, large enough that a single seed of the symmetric baseline returned a spurious -0.82 gap.

Each trader’s private value is $F_t + \varepsilon_{i,t}$ with $\varepsilon \sim \mathcal{N}(0, 5)$, redrawn each period. All experiments here use full information; the delayed and noisy signal treatments are implemented but not reported.

3.3 The decision contract

Code owns matching, budgets, order sizes and prices. An agent answers three typed questions and nothing else: direction (**buy/sell/pass**), urgency on a five-level scale, and whether the signal is already reflected in the book. A code-owned rule maps urgency to an integer tick, anchored on the book and hard-clamped at the agent’s own private value.

That clamp is load-bearing. *An agent cannot express a losing price*, because it never expresses a price. A 20,000-case property test asserts that no quote is ever on the wrong side of the quoting agent’s value. Consequently every effect we report is about direction and timing, never about arithmetic.

The question wording is frozen and hash-locked; a test fails if any instruction or criterion changes. This is the defence against unrecorded prompt iteration.

3.4 Arms

All four arms emit the same object, including a distribution over `{buy, sell, pass}`, which makes them comparable.

- **ZI**: random direction and urgency, fixed policy distribution, value-clamped so it never knowingly trades at a loss.
- **NBR**: logit best-response over the same urgency grid, scored by expected surplus times a fill probability. Exactly symmetric between sides by construction; a buy at mispricing $+x$ and a sell at $-x$ receive identical probability.
- **Jev-argmax**: the model’s modal action.
- **Jev-sample**: a draw from the model’s distribution.

The two model arms read one response, so for a given observation they are one API call and differ only in the decode rule.

4 The model and its measurement properties

The model is a commercially available typed-decision model that returns, for a multiple-choice question, a selected option together with a probability over all options. Three properties affect measurement and are reported because they constrain what can be claimed.

Non-determinism. Identical requests return different answers. Four repetitions of one request returned urgency scores of 2.84, 2.79, 2.82 and 2.74 and buy probabilities of 0.98, 0.99, 0.99, 0.99. There is no seed parameter. Run-to-run noise is roughly ± 0.05 on a 0–4 urgency scale and ± 0.01 on probabilities. *Reproducibility therefore requires the response archive, not merely the code*: a rerun without it does not reproduce. All 118,628 archived responses are retained and every reported figure is regenerable from them.

Quantisation. Probabilities are returned to two decimal places, so three-option distributions routinely sum to 0.99 or 1.01. We accept and renormalise within tolerance.

Undocumented failure modes. The published status codes are incomplete; the service returns bare 5xx from its edge under load. A client written to the documentation alone will treat these as fatal.

5 Experiment 1: wording, conviction, and a null

5.1 Domain wording induces a conviction asymmetry

Our first criteria were written in ordinary domain language: *buy* as “the good is underpriced... you want to acquire a unit”, *sell* as “the good is overpriced... you want to dispose of a unit”. At

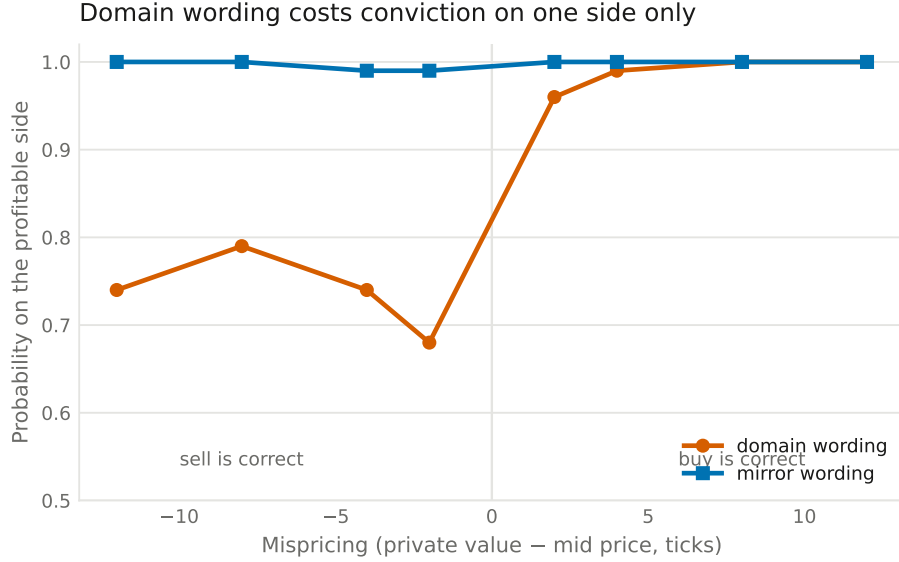


Figure 1: Probability the model places on the profitable side, against mispricing, under the two wordings. The model is symmetric when asked symmetrically. The effect is $25\times$ the model’s own run-to-run noise.

equal absolute mispricing the model placed 0.988 on the profitable side when buying was correct and 0.738 when selling was correct, a gap of 0.250, with the missing mass on *pass*.

The gap survives a second price level (+0.225), reversing the option order (+0.235), and renaming the options neutrally (+0.185). It does not survive rewriting the two criteria as exact structural mirrors—“value is *above* the price” versus “*below* the price”—under neutral instructions, where the gap is +0.005 (Figure 1).

Two points matter for practice. First, the modal action is *correct on both sides at every point tested*: a practitioner reading only the selected option sees nothing wrong. Second, the wording that induces the skew is the natural one; symmetry had to be engineered deliberately.

5.2 It does not move pricing error

We predicted this skew would slow price discovery after negative shocks. A four-seed exploratory run supported it (+0.63, four of four seeds positive, Holm $p = 0.061$). We registered the test and ran it on twenty fresh seeds. It failed: -0.184 , 95% CI $[-0.479, +0.111]$, seven of nineteen seeds positive, one-sided $p = 0.897$. The confirmatory interval excludes the exploratory estimate. With $SE = 0.140$ the run had ample power to detect the exploratory effect, so this is evidence of absence.

We report it as such. Wording skews what the model says; it does not, in this market, change what prices do.

6 Experiment 2: liquidity failure under modal decoding

The null above was correct for the question it asked, and the question was wrong. Root mean squared pricing error is defined only over periods in which trade occurred, and so discards exactly

Arm	Traded share		Gap
	after up	after down	
Jev-argmax	65.6%	91.3%	+25.8
Jev-sample	92.2%	95.3%	+3.1
ZI	76.0%	75.6%	−0.4
NBR	98.7%	99.7%	+1.0

Table 1: Preregistered confirmatory test, seeds 24–43. Levels and gaps are both computed from the same run; the gap column is the mean of per-seed gaps, which is why it need not equal the difference of the pooled levels exactly. D1: the Jev-argmax gap exceeds zero, $t = 6.37$, Holm $p < 10^{-5}$, eighteen of twenty seeds positive. D2: it exceeds the Jev-sample gap by 22.7 points, $t = 6.01$, Holm $p < 10^{-5}$.

Arm	Shock	No seller	No buyer	Both sides, no cross
Jev-argmax	up	100.0%	0.0%	0.0%
	down	96.2%	0.0%	3.8%
Jev-sample	up	71.4%	4.3%	24.3%
	down	57.1%	7.1%	35.7%
ZI	up	0.9%	2.8%	96.3%
	down	2.3%	1.8%	95.9%
NBR	up	8.3%	0.0%	91.7%
	down	0.0%	0.0%	100.0%

Table 2: Why each silent post-shock period was silent, twenty seeds. Under modal decoding the market is missing a counterparty essentially always; the symmetric baselines fall silent the ordinary way, with both sides quoting and prices failing to meet.

the periods where the effect lives. In one run, modal decoding produced zero trades in all three up-jump windows and full trading in all three down-jump windows—270 trades in the run, none in the 45 periods after a positive shock.

We therefore registered a new outcome, defined for every period: $\text{traded}(d)$, the share of the fifteen periods after a jump of direction d in which at least one trade occurred.

Mechanism (exploratory diagnostic). We instrument resting depth on both sides at the close of every period, which separates two failures: *no counterparty* (one side empty) and *no crossing* (both sides present, prices do not meet). Table 2 classifies every silent post-shock period.

The failure is one-sided: the market runs out of *sellers*. It is essentially never short of buyers (0% of argmax silences, 4–7% of sampling silences). This classification was added after the effect was found and is exploratory corroboration, not an independent confirmatory test; E1 is the preregistered mechanism prediction.

Unanimity is directional in the market. A static probe of the model, holding the book fixed at its pre-shock level, returns agreement of 92% after an up shock and 96% after a down shock. Taken at face value that inverts the result: more agreement on the side that halts less. We initially took this to mean unanimity could not account for the direction, and posited a

Period	After up shock			After down shock		
	Buy	Sell	Pass	Buy	Sell	Pass
t	90.0%	7.9%	2.1%	34.6%	59.4%	6.0%
$t + 1$	72.7%	23.1%	4.2%	51.0%	37.7%	11.2%
$t + 2$	72.1%	23.1%	4.8%	47.9%	40.4%	11.7%
$t + 3$	69.0%	25.8%	5.2%	47.5%	44.4%	8.1%
$t + 4$	67.5%	26.5%	6.0%	47.1%	45.0%	7.9%
window	65.9%	28.2%	5.8%	48.8%	42.3%	8.8%

Table 3: Jev-argmax order flow by period after a shock, twenty seeds. At the shock itself the arm is near-unanimous on the up side and takes the correct side on the down side. The sell majority is gone by $t + 1$, once the book has begun to move, so the fifteen-period window average understates what happens at the moment that matters. Reporting only the window would suggest the arm never reaches a sell majority, which is false.

second mechanism. That was wrong, and the error was using a probe statistic as though it were a market statistic. The probe holds the book frozen and sets the trader’s signal equal to its own private value; the market does neither.

Measured in the market, agreement is already directional (Table 3). In the first period after a shock, 90.0% of traders buy following an up shock, while after a down shock 59.4% sell against 34.6% buying. Unanimity after an up shock is near-total; after a down shock it is far weaker. That ordering matches the halt, and no second mechanism is required.

The effect therefore has a single mechanism. Unanimity is near-total after up shocks and materially weaker after down shocks, which sets both the magnitude of the halt and which shock it strikes; and it dissolves with market size exactly as Section 7 predicts. The sampling arm exhibits weaker correlation throughout, consistent with the size results.

What remains unexplained. Why the arm is more decisive on the buy side than the sell side at equal absolute mispricing—90.0% against 59.4% at the shock—is not established here. It is not reproduced by either symmetric baseline, and it is not removed by the mirror wording of Section 5, which eliminates the *stated* conviction asymmetry entirely. Candidate explanations we cannot separate with this design include the model itself, the option naming in the response schema, the under-specified referent of “the current market price” in the instructions, and the omission of inventory from the state. We flag it as the clearest target for follow-up.

A control that does not quite hold. NBR returns +1.0 points ($t = 3.33$), small but distinguishable from zero, indicating a slight genuine structural tilt in the market. It is 1/26 of the effect under test, ZI is clean (-0.4 , $t = -0.22$), and it does not survive the per-size controls of Section 7. We report it because the registration required controls to sit on zero and this one does not.

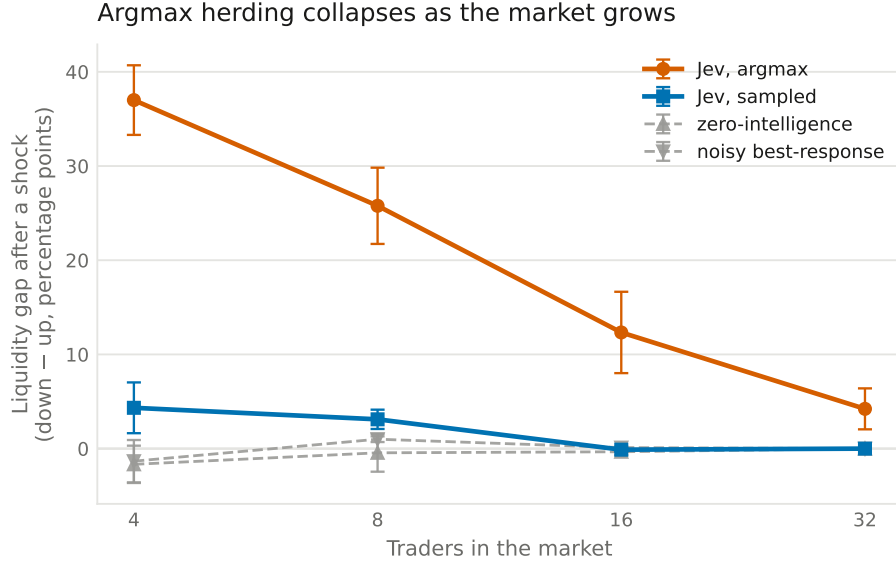


Figure 2: The liquidity gap against market size, twenty seeds per point, bars are standard errors. At $N = 32$ both symmetric baselines trade in 100% of periods with a zero gap, so there is no structural floor and the residual 4.2 points is attributable to the arm.

7 Experiment 3: the mechanism, tested where it could fail

If the market halts because every trader reaches the same conclusion, the effect must depend on the number of traders. Taking the agreement rate measured in the market at the shock, $p \approx 0.90$ after an up shock (Section 6; the static probe gives a similar 0.92, but we use the market figure because the probe is not a market statistic), the probability that all N agree is approximately p^N , falling from 0.66 at $N = 4$ to 0.03 at $N = 32$. We registered two predictions before running: that the argmax gap would decay with N (E1), and that the sampling gap would not (E2).

Statistics. The sweep reuses the same twenty seeds at every market size, so pooling all eighty observations into one regression would treat repeated measures as independent. We estimate a slope within each seed and test over seeds. We note also that the $N = 8$ argmax cell is the same data as D1.

E1 confirmed. The gap falls from 37.0 points at $N = 4$ to 4.2 at $N = 32$: slope -0.112 per doubling, seed-paired $t = -7.78$, nineteen of twenty seed slopes negative. (The pooled estimate, $t = -6.91$, understates it.) It is robust to dropping the $N = 4$ point, which we flagged in advance as floor-constrained because the symmetric baselines themselves trade in only 33% of periods there.

E2 not resolved. We predicted sampling would show no decay. The seed-paired estimate is -0.016 per doubling, $t = -2.00$, two-sided $p = 0.060$, with thirteen of twenty seed slopes negative. That does not falsify E2 at conventional levels. It also does not confirm it: no equivalence margin was preregistered, so a non-significant slope is not evidence of no slope. An earlier draft of this

paper reported E2 as falsified on the pooled $t = -2.53$; that was an artefact of ignoring the repeated-measures structure, and we withdraw it.

What can be said is weaker and still useful. Sampling’s gap is small at every size (4.3, 3.1, -0.1 , 0.0 points) and its slope is roughly a seventh of the argmax slope, though the ratio rests on a marginally estimated denominator. Sampling does not abolish correlation; whether it decays with size on the same curve is unresolved here.

8 Discussion

The practical claim is narrow and, we think, general in shape: *the best way to use one model is not the best way to use many of them*. Modal decoding selects the most probable action for each agent and is collectively degenerate when agents share information and require counterparties. Sampling selects a less probable action some of the time; we did not measure per-agent trading surplus, so we do not claim it is individually worse in payoff terms. Here it was the difference between a market that cleared and one that did not. We do not claim it is the only remedy—any source of decision diversity might serve.

The failure is largely not a reasoning failure. At the shock the modal action takes the correct side in both directions, and across our static probe grid it is correct at every mispricing. The loss occurs in aggregation, and evaluation of a single agent’s decisions cannot detect it. We do not claim the judgement is flawless: 34.6% of decisions at a negative shock are buys at a mispricing of roughly ten ticks, which is the residual discussed below.

9 Limitations

The asymmetry in initial agreement. Section 6 establishes that agreement at the shock is directional—90.0% take the same side after a positive shock against 59.4% after a negative one—and that this accounts for which shock the halt strikes. It does not establish why the arm is so much more decisive buying than selling at comparable mispricing.

Two candidate explanations are not reproduced. It does not appear in the symmetric baselines at any market size, though a market feature could in principle produce an asymmetry for one policy and not another. And it is not removed by the question wording. Under the mirror phrasing of Section 5, which eliminates the stated conviction asymmetry ($+0.005$ against $+0.250$), agreement at the shock is 89.8% after a positive shock against 64.8% selling after a negative one. Mirror wording is if anything *more* willing to sell at the shock than domain wording (64.8% against 59.4%), and the up-side figure is unmoved. Whatever produces the asymmetry survives making the question symmetric.

What remains is the model itself, the task framing, the option naming in the response schema, or the state representation, and we cannot separate these here. We flag it as the clearest target for follow-up work.

One model. All results are for a single typed-decision model. We have no evidence about others, and cannot say whether the agreement rate that drives the mechanism is characteristic or idiosyncratic.

Provenance of the probabilities. We treat the returned probabilities as the model’s beliefs. We have asked the vendor whether they are calibrated posterior mass or post-processed, and

have no answer. If the latter, Experiment 1 is a statement about a serving stack rather than a model.

Full information only. Every trader observes F_t exactly, which maximises the correlation the mechanism exploits. The delayed and noisy treatments are implemented but unreported, and would plausibly weaken the effect.

Scale. Markets of four to thirty-two traders over 160 periods. The extrapolation to larger markets is the fitted decay, not a measurement.

The estimand is memoised, and this narrows the claim. Requests are content-addressed: when two traders render the same tick-rounded state, the second receives the stored response rather than a fresh call. Because the service is non-deterministic, that is not the same system as N agents each calling it independently.

What we measure is therefore *one archived response per distinct state, reused across agents and periods*. Under modal decoding the two coincide more closely than one might expect—the rounding probe shows the model is decisive across the whole realistic range of private values, so independent calls would very often return the same mode anyway—but they are not identical, and memoisation can only increase within-state agreement. For the sampling arm the gap is larger: our traders draw independently from *one frozen distribution* per state, whereas independent callers would draw from distributions that themselves vary.

The consequence is a scoping one. The argmax result is evidence about memoised model agents under shared information; generalising it to every deployment of modal decoding requires a repeated-response condition we have not run. It also qualifies the sentence “a deterministic decoder makes them act identically”: a deterministic decoder returns identical actions only for identical input distributions, and memoisation is part of what makes the inputs identical here. A sensitivity analysis drawing several live responses per state is the right follow-up.

10 Reproducibility

Because the model is non-deterministic and has no seed, code alone does not reproduce these numbers. Every response is archived—118,628 of them—and the analysis can be replayed against the archive with live calls disabled, in which case a cache miss raises rather than silently contacting the API.

The archive holds 118,628 *unique* responses, one per distinct request. That is a lower bound on live calls rather than a count of them: retries, repeated uncached requests and overwrites of the same key would not appear separately. Estimated spend from the archived token counts is \$3.85, consistent with the provider’s reported balance. These figures, and SHA-256 hashes of every result file and archive bundle, are generated from disk into `data/manifest.json` rather than transcribed.

The preregistrations, including the predictions that failed and the reasons the outcome measure changed between Experiments 1 and 2, are in the repository with their commit timestamps.

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